

ActInf GuestStream 087.1 ~ John Boik "CogNarr Ecosystem: Facilitating G

GuestStream | John Boik | 1:43:01

All right, hello and welcome. This is Active Inference Guest Stream 87.1 on September 5th, 2024. We're here with John Boyk and Lee Ayer. So Lee, how about please say hello and give some contextualizing thoughts. Introduce John however you like and then we'll have presentation discussion with John. So go for it Lee, thank you. Sure, thanks Daniel. So I came across John through the Active Inference Institute because he was the only other person sitting on the strange intersection that I'm sitting on in this space between looking at societal transformation and active inference as well. So I was very excited and I joined the first live stream that he did on his science-driven transformation papers and subsequently I contacted him and we both explored a kind of a shared interest I guess in functional forms of cognition at scale and how those can be facilitated and John was talking about the potential of some kind of software solution to facilitate that and the kind of the discussion that ensued over the following months has led to John creating this paper. So I've kind of been in discussion with him sort of over the months and I'm really really interested to get more into this presentation. Okay thank you go for it John. Thanks Lee, appreciate that but I think it was more like many months of discussions that we were engaged in but led up to this paper. So thanks for all your input during that time. So my name is John Boyk. I'm a research fellow at the Active Inference Institute. I'm very pleased to be appointed to that position. I have two new preprints available to the public, two papers that we'll be discussing today both on the topic of group cognition at scale as Lee mentioned uh I'm I'm proposing a a essentially software ecosystem online ecosystem for group cognition at scale and uh and we'll today we'll talk about uh kind of the some of the general ideas there the overview of that project and then uh on the second guest stream but maybe is this slides view can everyone see the slides right now and can you go to the next slide then please

um

so there's two two papers have two guest streams that we're doing uh today and then the next one is on September 10th uh and on the second one today we'll cover some general life general concepts and on the second one we'll get a little bit more into some technical details about the underlying meaning representation uh of when we'll discuss in a moment what that term meaning representation means is but this first paper is available on archive you can download the paper from there and um and let's get started

next slide please

so i'm proposing something called the cognar ecosystem and cognar stands for cognitive narrative and in concept and i and i should say this is just a concept this is uh this project is still in its incubation phase uh um uh the the two papers are the substance of what has been done so far on this project and i'm hoping that i can interest others in the research community and in other communities civic civil society communities and other communities to get engaged with this project and help it to move forward so in concept this is an open source uh online extensible ecosystem of basically software but

resources tools applications library services that facilitates group cognition and especially for large groups um and and especially with regard to deliver deliberation strategizing problem solving uh decision making and that kind of things but you'll see that as we go through these slides you'll see that there's really a variety of use cases for this but these are the use cases that i really had in mind as i was um developing the concept for the for the project so essentially cognar helps users to what i what the way i phrase it is tell their story and by that i mean to convey their beliefs to the to what to their group about how the world works relative to some situation so um a group can be anything uh could be a you know could be a uh uh could be a a city could be a corporation could be a school could be any kind of group that is trying to you know process the the understandings and beliefs of its members towards some goal perhaps problem solving or some other goal so and towards that aim cognar helps users to tell their story that is convey their beliefs to the group about how the world works and by both i use the word story and i use the word narrative in the in the series series the series of two papers but i don't i don't mean to confuse the reader because i'm not talking about a narrative in the normal sense and like a written narrative where you write down several paragraphs or several pages of text it's this is a little this is a little different but i still call the representation a story because it's it's a story in the sense that uh that it conveys uh users beliefs about like why something happened and what might happen next and um what what should we do and what does it mean and all that kind of stuff so you know this this belief really a belief model that the user is is creating for this situation maybe explains you know what is the explains what the problem is what are we trying to solve or here's my solution or here's my solution or what just i think this is what just happened or what did happen or here's here's what i think will happen next um here's what i here's what i think will the people that will be affected by this issue uh i believe that so and so and so and so and so was involved uh i i think that this this situation means such and such and we should take you know to address it we could take these series of actions or i think this this this this this this or i think this should be done or or we could learn this from this situation so it's a story in in the sense that it it it spans the who what where how why questions regarding some situation next slide please regarding some situation next slide please so uh there's actually several sort of services and apps available for for for um group collaboration for example um uh voting apps and things like that um um this this the cognar ecosystem is i would say a much more sophisticated uh uh approach to that and um it allows a user to tell what i call what i refer to as a as a potentially rich story like that means the story could be long complex nuanced uh it could be conditional like for example i might believe that in some situation uh if if we do a and b then perhaps c will happen but if d happened first then i then something else is going on so it could be a conditional story uh it could be the could be uh the equivalent of maybe a few paragraphs of text very short a story or a story could be the equivalent of a book chapter you

know it all depends on what the user wants to do so so how could such rich information be processed by a large group

uh that's the really the question behind uh cogner how how could how could that happen so the proposal is for uh includes a back-end uh computational system that does numerous things it assists users and in creating and constructing stories it helps the group to digest what all of these stories are saying and how they are the same and how there's different and you know what interesting points you know other stories make and where there's agreement and you know what are the what are the many ways that this issue is seen by this group of people

um so it does like analytics on stories for example to help answer some of those questions

uh the system also assesses stories for quality like you know is this is this story uh uh on topic or whatever the group is trying to do whatever the whatever the situation is the group is addressing uh is that complete does it does the story just make sense in and of itself um and um and the the system can use the that those kinds of assessments to give feedback to the user like uh for example the a story might have some character in it named bob and some other character named sue and it might not be clear from what the user is creating whether how those two people are related so the system might if the system is confused about what the relationship is it might actually ask them like ask the user like what is i'm a little confused what's the relationship between these two people so in that sense the system is acts very much like a like an engaged active listeners is like if you were at home telling your story to a friend the system is kind of like that friend that is listening and trying to make sense of what you're saying and maybe asking you know pertinent questions to help direct your uh your telling of the story so that it understands what's what what the story is about and the system also performs inference for uh uh for queries like perhaps other users other people in the group have a question for the for the storyteller or the story or perhaps the system has questions like what's the relationship between bob and sue as i just mentioned so the system has to perform various kinds of uh inference to help generate and answer those types of queries without without such a computational system the the you know the the imagine imagine a large group of thousands of people and thousands of people submit a long complicated story how would a group make sense of that if there weren't some

kind of computation computational system doing assessments and inference and and guidance and assistance

kind of behind the scenes next slide

so i've been talking about a group uh a group that's interested in some some topic some situation or topic or problem but maybe we should uh take a moment and and just let me explain what what i mean by group so we're talking about a group of people in this case and um and i i call i call a group a collection of people that demonstrate some kind of coordinate coordinated behavior in response to dynamic conditions as as time elapses and conditions evolve so examples include just about anything you can think of that would be a group so social clubs volunteer groups civic organizations non-profits corporations and even political entities like cities and states and nations

a group can range from the very informal to formal flexible to structured temporary to permanent small to large autocratic to democratic and the way it manages information and manages decision making and so a group senses a group does many things you know generally a group has a purpose it forms for some purpose and to carry out its purpose it generally senses its world processes information anticipates outcomes the like the a group might say you know we we should take steps a and b in order to be financially viable or you know some such thing a group evaluates its options it makes decisions and then it takes actions either just within the group or within the world or in some way some kind of actions well in other words i'm saying that a group can be thought of as an organism an organism that performs cognition and just like any organism an animal a rabbit a human the animal has to do all of these things it has to sense it sense its world sense danger in particular look for food it has to process the information that it receives from the world and some of that information of course can be um uh you know noisy like it thinks it sees a wolf but maybe that's not a wolf what does it do what what should the animal do in this situation you know it has to anticipate the outcomes of its actions evaluate options make decisions and then take some kind of action in order to in order to maintain itself in the world and really in order to thrive in the world if that's possible so so i'm i'm thinking of a group as an thinking of a group as an organism that organism is just like any organism is performing cognition next slide

so that that's a quick overview of cogner but uh now you know maybe it's a good point just to ask the question you know who cares or more in particular why should you care the the listener um so um obviously cogner is about group cognition um and maybe you see the need for improved cognition maybe maybe in particular you wish that some group perhaps a political group a nation or a state or or a corporation or some other group maybe you have a story to tell that you think would be important to help this group out to help the people to protect to to warn of danger to highlight a problem or or a solution that you think no one else is thought of or is being underrepresented so maybe you have beliefs maybe you have a story to show share with a group and you want and not only that it's not just sharing a story because because if i i could write a a 50-page letter and send it to my congressman or you know some other some other group group or group member and there's really no guarantee that that group will really understand what i'm saying in this in this thing you know that's um so you so whatever you share with a group you want them to you want the group to actually hear it to register it to hear it and then understand what it is that you're saying because why bother if uh why bother submitting information of if it if it's not really heard or understood or maybe you maybe you look at the world as as i do and you see that the world has some serious uh you know the human world has some serious issues to solve climate change should be one example but there's many pollution climate change poverty you know all kinds of social problems and uh and maybe uh maybe you feel that you know isn't there some way that we could do better isn't there some way that as a group says as small groups as large groups as formal groups as informal groups is is there some way that we could cognate better that we could be more functional organisms group organisms and better act better make better decisions in our world and take better actions you know maybe maybe you think we could do better or maybe you just are a scientist

and you're interested just simply in the science behind cogner and and i would say that this project is large uh it's and complex and it touches or or could touch really just about every field of science uh i mean if you give me a a scientific field i could take a a guess as to how this cogner project might affect it or how it incorporates ideas from that field let's just take one example say medicine like you might think like what what what what what could what would cogner have to do with medicine because it's not a medical program you know in and of itself but doctors have to make decisions groups groups of doctors insurance companies other groups have to make decisions so that they have to come together as a group to make decisions and and assess some situation so that's one potential way another another potential impact on medicine is as a patient i have a story to tell i have a story about what happened you know there was an accident i i cut my arm the i stepped on a nail it was rusty this and this you know this and this happened and i'm i'm afraid of this and this and i have a condition past condition of this and this and i'm allergic to this and this and all of that can just be written on a piece of paper you know is it like a regular intake form but but um in cogner that the story is standardized in a sense so that it's easily readable and understandable by humans and computers and your a story that a patient is telling you using a the cognitive system would be um more accessible to inference and computation and um and there would be a and it also there's a natural record of of stories in cogner so there would be potentially many areas of medicine as just one example of uh of how either uh cogner incorporates the science or could affect the science um or maybe you're just curious about uh you know what's possible for human groups you know we we live in a society that uh is faces all kinds of challenges uh there's future challenges on the horizon uh you know what what is the potential of society how how is considering that a society is a group of people so again we're talking about groups how how how could how could a society best share information make decisions uh cognate really how could the organism called a society perform functional cognition or or or improved cognition over what is common or available today so i i i hope there's many reasons why you might care about cognitive uh next slide please so i've mentioned the term large group already so what you know what does large group mean what does it mean to me uh dozens question mark hundreds thousands millions of participants and uh i would say the answer is yes all of those right um maybe dozens of people perhaps can fit in a room excuse me fit in a room and and talk freely about some situation some topic and uh and share rich information verbally or you know in other ways face to face that might work with dozens but it doesn't work so well with hundreds uh and it certainly doesn't work so well with thousands or millions so um as i said a moment ago cogner has many potential use cases but but one that i'm really targeting is the large group setting where really you know hundreds to thousands or millions of people are um are involved in the group and and having said that there's also use cases for just a single person using cogner there would be reasons why somebody might want to record their story in card now just for personal reasons to as a history for example to look back on later and reflect when and there's other reasons too that we'll talk about shortly so so the question at hand is you know what kind of online ecosystem would allow say a hundred thousand people to share some kind of rich uh complex story and and in particular a dynamic story like i i i'm a member of a group we're sharing our stories i i write down

what my story is i look at what you're writing i look at your story and i go oh that's a i like that idea i like that part of it or that's that seems really useful to me or oh i forgot i forgot yeah that i should have mentioned that myself so by sharing stories by sharing belief models we can come to you know uh reach a you know better agreement our our belief by sharing our belief models can start to um influence one another and and develop and the group then develops a shared belief model just to an extent um and then and then it's not just about cagnar is not just about sharing uh stories it's about using them in some kind of uh to facilitate functional uh group cognition right so you can imagine a group a corporation or any kind of group civic group uh a city state and the cognition for that group you might say is functional sometimes or really quite dysfunctional sometimes so cagnar is aiming at facilitating functional group cognition that's its purpose next next slide

okay so um we've talked about what a group is uh and i've not and i've used the word cognition now many times so maybe we should talk about what is cognition um so in cognitive science one might come up with various definitions of you know what is cognition how does it work what is its purpose uh so i'm relying quite heavily on the active inference uh uh framework for the you know my definition of cognition and the way the cognitive system would work but uh active inferences is uh is essentially a bayesian description of cognition it's a normative principle uh underwriting cognition and biological and artificial agents so in active inference there's an intelligent agent and a world that the intelligent agent is acting in maybe maybe there's even multiple intelligent agents acting in the world and an agent has an internal generative model about how the world works uh that is it it's given some conditions um it sees that the that there's potential actions that could be taken and it and it sees that there's likely outcomes of various actions so uh an agent has a model of of how the world works what would have what will happen if i do a what will happen if i do b what will happen if i don't run from that lion um and uh the the the the the the the agent and its world are separated in a sense statistically separated by what's called a markov blanket and that blanket uh consists of agent actions and agent observations so as an agent i take some action that action affects the world the world reacts in some way according to the world's generative process and then things change conditions change and as an agent i perceive my world but you know maybe it's dark and i don't perceive it very closely or maybe it's loud and i don't hear it very accurately or you know i get sensations from the world but they're not necessarily i'm not necessarily observing the actual state of the world i'm observing some kind of noisy representation of that so uh so the the act the markov blanket is a it's it's it's conditionally separates the world from the agent and active inference is the framework that i'm using to to talk about cognition uh and and and not just cognition how it works but also what is the purpose of cognition what is what is functional cognition versus dysfunctional cognition next slide please so what is the purpose of cognition well from an active inference perspective it's we can think of it you know simply as an idealized four part uh four part cyclic process of i i predict what will happen if i act i i choose an action from my generative model i i s the world changes i sense whatever however the world has changed and i learn from processing those sense sense inputs for example uh

suppose i'm a young boy and i'm learning how to catch a baseball right if i don't lift my arm high enough i maybe missed the ball so that happens a few times i i learn that i really need to lift my arm higher and if i want to catch that ball so uh so you know again uh informally and simply it's the cycle of predict act sense and learn and an agent does this it engages in cognition to reduce its uncertainty be uncertainty about whether it's going to maintain achieve and maintain those conditions that it's thinks will be good for it that will that will help its thriving as a as a young boy learning how to catch a ball i i want to catch the ball that will be good for me if i can learn how to catch a baseball and i want to i want to reduce my uncertainty that i'm going to catch that ball i'm going to practice hard and i'm going to learn from my mistakes and i'm going to reduce the uncertainty that when i put my glove up that ball is going to land on it or or i'm going to outsmart the lion and get and get away from danger when the line appears uh so an agent favors those choices those actions that are and going to that are going to ensure the greatest resolution of uncertainty under the constraint that preferred outcomes are realized that's the way friston phrases it so so you know so you know it's it's it's it's not just that i want to learn how to catch a ball or run from a line it's that i want to be sure or pretty sure that i can catch that ball or or get out of the way of that line and uh and from all this i would say that active inference offers a potential description of functional versus dysfunctional cognition if if i have a pattern of making poor predictions and therefore acting in ways that are that are harmful to me i you we could call that dysfunctional cognition if i fail to remember what happened last or fail to remember how many times or the many ways that i missed the ball that could be also called dysfunctional cognition and cognition itself rests on many core functionalities like remembering and attention and and uh and communication and other other you know processes and any one of those or all of them together and any one input in individually could be functional or dysfunctional in some you know cognitive event uh next slide please so so given all of that background what is the cognar ecosystem really so i'm i'm gonna i'm gonna i'm gonna claim or offer that that cognar could be viewed as a cognitive architecture so every organism has some kind of every organism cognates whether it's a amoeba or a human and um cognition cognition requires some sort of cognitive architecture uh for humans we have a central nervous system and and we also have computers and external tools and things that can be thought of as part of that cognitive architecture and uh the cognitive architecture essentially integrates information and and helps us coordinate our actions in the world and it's the and it's the set of components by it by which and through which cognition occurs and for a group that includes uh the a group has members so it includes the cognitive architecture of the individual members but it also includes the this set of norms that a group uses its rules its mechanisms and institutions its tools its its sensors perhaps we're talking about a group that is a nation and it has sensors on air pollution you know something like that uh it's analytic processes its systems procedures all of which uh facilitate or guide cognition really make cognition possible and every part of a cognitive architecture can be functional or dysfunctional right so for a group say um maybe it's rules maybe it has certain rules that that are either enhance functional cognition or or detract from the functional cognition so i i view

cognar as a component of the cognitive architecture architecture of the groups that are using it
and again its purpose is to facilitate functional cognition versus dysfunctional cognition
uh next slide oh lee yeah

yeah hi john um sorry this is really interesting because um you know but based on the discussions
we've been having uh i'm kind of using the concept of cognitive architecture and some of the research that
i'm doing in organizations at the at the minute and um and tried to feed it back in so if you take a
an organization like a public sector body or a a commercial organization um would you would you
consider that the the the structure of the processes are all part of the the cognitive
cognitive architecture of that particular entity well i mean i suppose depending on how broadly you want
to define a an architecture i suppose it could include just about every aspect of a group you know
potentially but i think in a practical sense you could probably focus on those components of group
you know those mechanisms and structures of the group and its dynamics that would be particularly aimed at
uh sensing the really the four steps that i uh discussed just a minute ago uh taking action learning uh
you know sensing the world uh you know one of the things i guess one of the reasons i ask is is that
one of the things that uh tends to uh the the context and organization has a an enormous influence in
shaping people's behavior um but uh because there's a there's a general kind of widespread view that
cognition is localized within people rather than a kind of uh you know an organismic view of it i guess
where it's kind of distributed between people um in a hierarchical organization a lot of those factors
if i mean i'm almost viewing it now as a generative model but there's no way that certain processes
that determine behavior can actually be subject to change because they're not considered just part of
the cognitive structure if you if you see what i mean so i think the idea that um you know that
that um groups could view themselves as organisms is really interesting but currently there's kind of
more of a mechanistic kind of view of of organizations and i think that potentially gets in the way anyway
maybe this is a discussion for a for a different no no no no no no no it's actually good because i i i i i
i wanted to emphasize the importance of this view you know thinking of a group as an organism as a
cognitive

organism and thinking of decision making and and those operations as aspects of group cognition
so you know uh the world is kind of crazy in some ways politics are engaging for many people
and this is just one example in the in the for a political entity but you know we we
we rarely ask what a uh what a economic system is supposed to do or what a governance system is
supposed to do or or really what an organization is supposed to do right so so it what we're framing
all of that now in terms of cognition and we're framing it in such a way that it is amenable to
scientific you know examination of whether this is functional or dysfunctional and and we're and and
from an active inference perspective cognition has a purpose so so then we can start talking about is this
organization that's the you know it's structures that the way it the way it's
put together its rules its mechanisms its institutions is that functional yeah and and
right and could it be improved like now that we know what we're after now that we have the problem
you know the the issue and focus yeah how could we improve the the cognition of of uh

uh i can just give you a tiny little tiny little vignette so in this particular organization they have a performance management system uh but the so that if you if you're to take the view of functional cognition you know the purpose of the you know what's the purpose of the performance management system what's there to motivate employees reward them you know keep them feel excited about their work but it doesn't do any of that what it actually does is it demoralizes them annoys them gets in their way and just generally pisses them off i mean it's actually doing the opposite of what's intended you take the whole system out and the organization would be better off but there's no it's not subject to any prediction error signaling because it can't be right no defined it in terms of you know purpose and outcome so it's a completely dysfunctional system that's just stuck there that you can't take out and nobody you know nobody has a view on it it's it's really weird but i think would really get to it but anyway sorry to interrupt no no no no no it's good it's good i i just want to emphasize again that this is a in a sense it's a change in world view it's a it's this is a comprehensive and potentially very useful way to con to understand what a group is whether that group is a small informal group or a large formal group and and and then to assess you know how well it's doing and how could how could we how could it do better right so this that's the question for cognara that i raised just a minute ago is like okay given the setup given our goals given the purpose of cognar so what kind how should cognar be designed you know like how should it how should it be designed to do what we're saying it should do you know that's the question here um but anyway but getting back to the back to where we're at in the slides uh cognar is i i would say carder is a part of the cognitive architecture of the group that's using it next slide so actually this this slide speaks lee to some of the some of the things that you were raising there and and it speaks to some of the um some of the things that i just mentioned but so consider a society as a group say a nation as a group uh again a group that is that is an organism performing cognition for some purpose and that purpose obviously for any organism is to thrive to to maintain viability to to do well to live pass on information um so in the i wrote a series of uh uh three papers a couple of years ago in 2020 and 21 2021 and i think maybe there's i don't know daniel four guest streams for that series i think three or four or something like that that are in the archives so i encourage viewers to check those out too um and that series of three papers again looked at uh cognitive architecture only in this case it looked at uh societal systems as the cognitive architecture of a society so by core societal systems i mean for example economic systems financial systems governance systems legal systems educational systems you know these constitute uh uh a considerable aspect of the cognitive architecture for society we use these systems to sense the world to think to make decisions to allocate resources to take a variety of actions uh that that hopefully actions that better the you know the nation at hand so um societies around the world the nations around the world have i should really say nations around the world have some kind of the core societal systems say economic systems that have some kind of particular design to them for financial system has a particular design but the question i raise in this paper is in the series of papers is out of all conceivable designs you know the existing designs or completely new designs which which of them might be most fit for this

purpose of group cognition which of these might best serve the common good and and obviously making good decisions that you know favor the vitality and well-being of a nation is serving the common good so out of any conceivable design or economic systems you know what is money for example is there some other way to think about money or some other alternative to money or how do votes happen or you know like any

aspect of a core societal system one could imagine many conceivable variations on the on a theme or just completely new designs new new ways of looking at what an economic system is and from this umbrella of considering all of these systems as part of a part of a cognitive architecture then one can then one can ask you know really what designs are fit for purpose and how do we of course how do we measure fitness and

you know what kind of information goes into assessing the fitness of some system but it opens the door for a science-based design of the systems by which societies live under and that is not the way that is not the you know our current system economic system financial systems governance systems they were not you know

developed through some kind of science-based process they were they kind of evolved through social social pressures and social conditions and and you know ideas from individuals and but we could revisit all of that the things that um and and ask the question you know what what what designs actually would serve us best and in this series i propose an r d program to really to address that very question not just not just you know the not just as an academic exercise of you know let's write some interesting papers about you know how how governance systems could be redesigned the idea in this series is how do you what kind of

designs would potentially be most useful and then how do you test them actually practically test them implement them use them you know gain information from you know the the experience of using them and learning and learn from them how how how could that happen how could that actually be done how could some kind of transformation you know i call it transformation in the series but how could some kind of transformational step like that occur and be viable and practical so in the in the series i offer you know what i think is one path towards that but this current uh uh series of two papers that we're discussing today this is an it's an extension of the same idea it's again looking at the cognitive architecture of a group and asking the question uh in this case for cogner what kind of design of cogner would could best help uh group cognition and how could cogner be used by groups to you know to help them perform better cognition next next slide

yeah you know now that i've said next slide but there's just one more thing i want to say about the last slide uh uh you know as i was about to say a few minutes ago lots of people are interested in politics and you know what a government should do and and they have ideas for uh you know like maybe somebody thinks that uh that um term limits on uh for judges or for politicians should be limited that you know that that would be a so you know that might be a solution to some of our problems or you know there's all kinds of of proposals floating around for how to improve economic systems or governance systems or you know any any of the course societal systems and and even say corporations do the same thing you

know they wonder how could we improve our communications and all that kind of stuff but the this concept this this this really powerful approach of viewing systems as the as a cognitive architecture allows those questions to become you know meteor in a way you know more relevant more with a greater high level understanding of what that system is actually supposed to do so i i believe that by viewing the the societal systems of a society as a cognitive architect architecture and looking at groups as as you know small or large groups as uh as as organisms that perform cognition i think that opens up all kinds of doors and avenues for investigation into what kinds of designs what kinds of you know rules and and approaches and methods and tools might actually help groups perform better cognition and better solve their problems okay next slide

so um everything we've talked about so far has just been sort of high level what is cognar what does it do what is its purpose but uh this slide really gets more to the to the question of how does cognitive work work under the hood so i i've already said that i use the word story but i don't mean story in the regular sort of narrative written narrative sense uh a user creates a story excuse me creates a story but and and it creates a story may perhaps by speaking you know in one version of cognar maybe a user speaks the story and the system hears it and understands it but maybe the first version of cognar would be somebody the user inputs a story kind of passage by passage you know sentence by sentence and um the system takes that input those little passages and other metadata associated with passages and constructs uh a fragment of uh of a of a story and that story is representative representative represented as a graph so it has its internal meaning representation representation is called a it's actually called a story graph and that story graph the reason that it's not stored in just regular the english text or text or some other language is that this representation allows uh is is more amenable to computation and could be better understood in the large group setting that in that sense there's other reasons too we'll get to but but the important thing for this slide is that is that actually what the user is creating is something that resembles a graph a graph of how little elements of the story and are are are are are related and story graphs are really the core innovation on which cardinar is based and that it is that representation allows cognar to curate understand reason with perform inference for uh stories and these stories then represent really high quality information uh much of the inform much of the vagueness of natural language is eliminated in this way or not all the vagueness but some of the vagueness

is eliminated in this way so uh at the top of this little graph you can see that a user types in for example text text passages those are translated by the system into a what i call a story graph fragment just a small little graph for that passage and that story graph fragment is then merged with the growing larger story graph and once it's complete that complete the complete story graph can then be translated and used in various ways it could be translated back into natural language so that potentially a user could listen to their story back like you know i i typed in this i typed in my story about you know what should happen for this particular situation and uh i could ask the system to replay that back to me and you can hear it perhaps uh it also this the story graph can also be translated or used as input into a variety of inference models so probabilistic ecological or other kinds of models and maybe it would be

[illegible]

it'll have such and such an effect.

These are all anticipations, predictions,
and the meaning representation has to be sophisticated enough
to handle all of those.

And they're not just predictions,
they're probabilistic predictions.

No one knows what's gonna happen in the future.

So all of those kinds of statements are probabilistic.

So even, you know, it has to be even more,
the meaning representation has to be sophisticated enough
to handle probabilistic predictions
about, you know, what a story means
and what might happen next.

So this is a very simple graphical version,
just to give you, the user, a very, very high level,
rough idea of what's going on under the hood.

But in actuality, the Cognar meaning representation
would be much more sophisticated.

Next slide.

So what is the purpose of the story graph?

Some of this I've mentioned already,
but it represents a story in a format
that is less ambiguous than natural language.

Natural language can be very ambiguous.

Very, the story graph is meant to be readable
and understandable, both by humans and computers.

So a human can look at a story graph or pieces of a story graph,
views of a story graph,
and verify that the system understood the story correctly,
for example.

But computers then can also read and process the stories
and then, you know, perform analytics and inference
and do the many things that the backend,
the computational system has to do.

The stories obviously then provide a structure
for inference and analysis.

And they also serve as interlingua,
to, you know, natural language independent representation

that's suitable for translations into other languages
or uses inputs to analytical models or other things.

So that's what a story graph should do.

And then one of the questions is, well, what does the Cognar story graph,
you know, what is its design?

You know, what is the, what are the specifics of its design?

And that is a topic that I raise in the, in the second paper
that we'll talk about a little bit in the next guest room.

Next slide, please.

So, we've talked about stories
and I've said a little bit about what they are,
but let's maybe recap that just a bit.

So, you know, a user has some story in their head about a situation,
some, you know, issue or problem in the world.

And, and the user has an internal belief model
about how the world works relative to that situation or problem.

And a story is a, is a, is an externalization of, of that belief model.

So, and that, and it, and it potentially is a dynamic externalization.

You know, it can change as the user sees other stories
or reads what other people think.

So it, it, it externalizes a user's belief system and makes it shareable.

You know, that's the purpose of Cognar is to share these belief models
with other members of the group.

So, again, it, the story conveys a user's beliefs
and the, the emphasis here is on beliefs, right?

Because the, the computer, the computational system
is not going to predict what's going to happen in real life
if the group takes actions A versus action B, for example.

And it's not going to, the system is not going to verify
that somebody's story is, is accurate, is realistic, or, or is practical
for the situation at hand.

Those are, those are assessments that the group of users has to do,
you know, together.

And the system helps that process, but it doesn't perform
those kinds of assessments.

So, so the system, the system doesn't have to know
all of the rules of gravity or, you know, Newtonian mechanics,
or it doesn't have to predict what's going to happen in the world.

It, it re all the system needs to do is to capture a user's belief model and to communicate that model with others and to help and others help the group understand that model and how it compares to other models and what elements are in it and, you know, what is its time span and, you know, what, what, what is, what, what is this, you know, what are the characteristics of this user's belief system and, and what is the user predicting will happen? So it's, it's the story just captures the user's beliefs, right?

And then the group shares and acts in some way on those beliefs.

So, so that's it.

The, the system helps a group to understand what its members believe.

That's, that's really the core of this.

And then those by sharing those beliefs, users can influence one another, for, you know, like in a decision-making setting.

Next slide, please.

All right.

So, so we've talked about what a story is, how it's representative and under the hood, what is, what the purpose of Cogner is, what the purpose of a story is.

And now we can talk about how this Cogner system might be actually used.

And I've said that the emphasis is on, you know, the cognition in the large group setting and particularly for things like decision-making and, and collaboration and things like that.

But in actuality, there's a very wide span of use cases that are possible with this kind of ecosystem.

And, and I'll cover just a few possibilities, but I, but these are non, this is a non-exhaustive list.

There's many, many possibilities and the communications in, in its different uses, you know, that maybe, maybe there's an app developed for each particular kind of use, but so maybe there's many apps that are part of the ecosystem, but the, but this, but the sort of the events, a, I call it a group, a group, a group defines an event and then invites people to participate in the event.

So for example, a city might, if a city is using the system and might define a group event to solve some particular issue, maybe a trash issue or a recycling issue or a road issue or some other issue and the, and the group would define the topic for the event and the purpose of the event and the rules for the event and the methods for the event and how decisions we made and, you know, the setup for how this event is going to unfold.

And then it asks for a group input through story graphs.

So depending on the event and the group and the purpose of the event, the communications

can be many to many as in, you know, collaborative decision-making, for example, that could be many to one or even one to many.

So, and, and, and, and I've been emphasizing group use, but again, an individual might use this just for their own personal use, just to record their, their understandings of the world at a certain time, for example.

So some possible use cases are in decision-making such as by a corporation or decision, corporation or a city, maybe customer service, emergency reports, public polling, instructions for how to put together bookshelves or instructions for how to run a machine, for example, system descriptions of, you know, what, what is, what is the system and how does it work?

Or maybe there's a whole variety of educational use cases also.

And later versions of Cognar could be, could include things like role-playing or game-playing, like intentional games, purposeful games and strategizing.

What, what, what, here's, here's my belief system of the world relative to the situation.

And you know, the people involved in this story or the characters involved in this story or, you know, so-and-so and so-and-so and so-and-so.

And what, what would happen if I, if I take action A, you know, what's likely to be the outcome?

What is, what is player B gonna likely to do if I take action A?

So later, more sophisticated versions of Cognar might include that kind of stuff where, as your user might play against the system or two systems in a sense might play against each other to, to, to, to, to, you know, to, to strategize for some situation.

Um, next please.

So the next couple of slides are just a few examples that I have in the paper and, um, for use cases.

So I'll cover them briefly.

Um, um, so I have written at the top of these slides, some text, and this isn't actually the text that's, you know, somebody would, uh, write in a, uh, in the real Cognar.

This is just a snippet, an example snippet of the kind of information that a user might want to convey, right?

So this user, there's an issue, some issue X and this, this, this person, uh, believes that in order to address that, we should first address issue Y that caused X and Y started when, you know, it's when such and such happened and it had the effect of such and such.

And this person believes that if we don't solve issue Y, then some event, a someone wanted event a is likely to occur maybe within a few months.

And then X, the original issue is going to be impossible to solve.

So, uh, if that, if a happens, then, you know, there's going to be all these effects maybe, you know, and there's some, this, this, in this story, there's some kind of, uh, you know, uh, 20% value put on the risk of this or, or the, or the, uh, outcome of this.

And, uh, this user says a possible solution to this, this problem Y is to do, you know, such and such, uh, and it's conditional.

So do such and such if event B occurs and such and such event C occurs.

So there's some kind of, you know, complicated story, conditional story involved here that has predictions in it.

Um, uh, it's, it's, uh, you know, there's some shoulds in it, you know, we should do this.

Uh, and how does, how does, how does all that work, uh, for the decision-making case?

Well, one common, I believe what would be a common setup for the decision-making case is, um, uh, a group forms or comes together for an event and, uh, individuals, you know, create their story.

Uh, uh, that's kind of at the top of this diagram, they create their story and then, um, you know, submit their story and the system, the middle, the middle block there, the system does its assessments and computations

and, you know, the various things that the system is going to do.

And then it reports back to the group about the range of stories that have been submitted and what they include and who they involve and, you know, what the kinds of, of solutions are, have been proposed and the various kinds of, uh, summaries and details about, uh, about the, the set of stories that have been involved.

And then the group, uh, uh, you know, uh, digest that information and looks at individuals can look at individual other, the stories of other individuals.

Uh, so users can kind of take in what the group is saying and maybe a user thinks, oh, that's, uh, that's, so, you know, uh, Bob had something really interesting to say there.

And I, I, you know, I should really include that as part of my story.

Uh, that was a good idea.

I didn't include it, so I should include it.

So, um, the next step is, uh, this, this process, uh, cycles in a series of rounds.

And at the end of each round, there's a duck kind of a dialogue and digestion phase where everyone looks at what's been done and gets a better idea of what the group is thinking.

And maybe they adjust their, their story, uh, you know, accordingly.

And hopefully over time, over a series of rounds, the, a proposed solution that the group favors starts to become clear.

Right.

And then, uh, then this process can end when it's sufficiently clear and there's enough, uh, agreement that we, we understand each other.

We, we think we understand the situation and we think as a group that this, you know, this, this, this solution or this set of solutions might be the best actions to take.

Uh, so that's the, that's group decision-making might occur over a series of rounds.

Uh, next slide.

Here's a, an example use case for emergency reports.

This is a, this is a person who wants to report that they just witnessed an accident.

Uh, there's a, there was an accident with some kind of truck of some kind.

Uh, maybe it was going too fast.

Maybe it was because of the wet conditions.

It shouldn't have been going so fast.

Uh, lots of accidents like side information.

There's been many accidents at that corner.

So maybe that, that maybe the, the, uh, maybe the local conditions have something to do with this.

But anyway, the truck had a telephone pole and now there's sparks in the grass and, uh, there's a school and children are up playing outdoors.

And, um, you know, I, I, all of this is a little unclear to me.

I'm about a block away, but, um, you know, that's what I see and I report it.

So, you know, there's a, there's an interesting story and you could imagine even a more complex story of someone that wants to report something.

Maybe they want to report a crime, for example, or in this case, an emergency report.

And there's also, there's various kinds of inference that has to happen to be like under the hood for this, right?

Like if you read the sentences that I, that I wrote there, I never say that a, uh, wire, uh, fell from the lines and is now on the grass.

That's sort of inferred, right?

So there's various things that the, that the, the system has to infer about the story in order to make sense of it.

And I say that there's a, you know, this user is saying that there's children present, but the implication is, is that there's a danger to those children.

Right?

So there there's actually a forecast, like a prediction, like if we don't do something, someone might get hurt.

So these children might be harmed, you know, because of the, the, uh, you know, the aftermath of this accident.

So, uh, so, uh, uh, you know, this is, this is, this is, uh, this little story is a lot different than, for example, calling a hotline and the hotline instructs you to press one.

If there's a fire press two, if there's a, you know, something else.

And then if you press one for a fire, it says, uh, you know, press one, if, um, the fire is in your house, press two, if you know, whatever.

So there are other computational systems out in the world that seek to address, seek to gather this kind of information for, you know, for, for the purpose of making an emergency report.

But I'm claiming that out of all, uh, out of all the kinds of systems that I know of, the Cognar system is much more functional and has the capacity to convey a much richer, uh, story understanding of someone's understanding about, about a situation.

Um, and perform inference on it and record as a history, the story and how it evolved and many other things that the, that the, uh, computational system does offers.

Next slide, please.

Um, this is a, uh, this is an example for customer service.

This is an example of, uh, of a, uh, of a, uh, a many to one sort of communication, right?

In the sense of, um, many customers are, are communicating with one agent, one, one central agent, which is the corporation in this case.

So, uh, uh, people are, are writing in and they're sending in their broken parts and they're, you know, they're, they're giving a description of why this part worked or, you know, is this part defective?

Or why it was this, why was this aspect of the part designs seemingly so poorly, or could you fix this design or, you know, like, here's what, here's what's.

Here's where it's broken.

This, this part here, this, this part F has a leak in it and it causes other part to H to slip, you know, some kind of sliding mechanism to slip.

And, uh, and, uh, and, you know, seems to me that part was, uh, it wasn't really broken.

It's just that it didn't really fit.

So it was that by design or is this part defective or, you know, is what's, what's, what's going on here with this.

So, you know, this is rich information that, uh, that a corporation might receive about, uh, its products from users and, uh, you know, much more useful than say, just, uh, uh, you know, a couple of sentences written in natural language.

Or maybe a paragraph written in natural language, because, you know, the corporation could use this to really assess, you know, really analyze.

Closely.

Uh, what, what, what's going on and how many times this has happened and how many times have people complained that that part was too thin and, you know, all that kind of stuff.

So, uh, there's, there's an example of, uh, of, of a, of a many to one sort of communication.

So, uh, you know, lots of political things happening in the world right now, elections and so on, and lots of polling going on too, for the, for the different elections.

And, um, you know, typically in polls, you might be asked a yes or no question.

Do you know, do you agree with the statement?

Yes or no.

Or maybe you have a, a, a short, um, uh, uh, multiple choice, you know, do you, yes, I agree strongly or no, I disagree strongly, or I can rate my agreement on a scale of one to five or, you know, some, something like that.

Um, but that is not near as rich as, as in taking what the user actually has to say and what, what they, what they think are the important aspects to, to this, to this, the question that's being pulled about.

In this case, the user is, is, is, is there's a poll for some candidate and, uh, this user is saying, you know, I thought, uh, her past decisions on certain issues are terrible.

And I've been fuming about them.

And, um, I don't trust her on this issue because of such and such statement.

And I don't even think she's sincere on that issue.

And anyways, that issue could be solved by some other much simpler process, which, you know, I, I, I

mentioned here and, uh, I don't think I'm going to vote for her.

So, you know, this is rich information.

And again, there's inference that has to go on behind the scenes with the, with the, uh, system.

Uh, for example, the word fuming, you know, that has various meanings.

Uh, does it mean that the person, the writer is on fire?

Uh, no, not in this case, you know, it means that they're angry.

So a lot of the various definitions of the word fuming, the system has to figure out what definition is the user aiming for here.

Um, next slide.

And then I think we're done with the next slide.

Uh, we're, we come to the end of our talk.

So, uh, what is next for this project?

Uh, I, I, I've already mentioned that Cogner is in the incubation phase and, um,

um, and I put out the vision in these two papers, vision for it.

And, um, really it's a big project.

It touches on many aspects of science, computational science, complex system science, all kinds of different sciences that touches on.

Um, and this is all beyond my capacity as one person to create.

So if this project is going to move forward, it, it will need your help.

Uh, whether you're a scientist or you're, uh, belong to a civic organization or you're just an interested citizen or whatever.

There's, there's, there's some way that, uh, just about anyone could be involved in this.

Um, especially as it picks up steam and, and moves, starts to move forward.

Um, there's many, many interesting questions to address.

I, I, I raise a few, uh, future research questions in the papers, but there's many, many more that could be imagined.

And I encourage people to reach out to me if they have interest in this project.

Um, upcoming, there's the guest stream on September 10th.

That's next Tuesday and, um, uh, I, I, I believe I'll be starting a Cognar project at the active inference Institute.

Uh, as people can get that involved that way and get involved with the Institute that way.

And, uh, they sh there should be some additional talks coming up.

There's a active inference symposium coming up that I think I'll present at.

And, um, and there's a few other talks I'll be giving to, to private groups and such.

Um, so, uh, again, please reach out to me if you have interest in this and, uh, and, uh, show up next time for the, for the guest stream on September 10th.

And we'll continue the discussion.

Awesome.

Thank you, John.

I will read some of the questions from Lockchat.

Upcycle Club wrote, is group cohesion one of the goals of Cognar?

Um, so good question.

And, and.

You know, the, the, in any kind of group, there may be very different opinions of what should happen.

And, um, there may be lots of friction in that group.

Right.

So, um, it's not that Cognar is, is necessarily going to solve all kinds of friction problems and get everybody on the same page for every setting.

You know, that would, that's not the way the world works.

Right.

But, but what it, what Cognar does do is allow, uh, sets up a process and a means and a mechanism by which people can share their stories and perhaps come to some agreement.

Uh, shared understanding, you know, share, share their way, they see the world.

And maybe the group will come to some kind of shared understanding.

If the purpose is decision-making, you know, that's kind of the context.

Um, is this kind of the interest in this group is known to make this kind of interest in this group?

But, but even if they don't Cognar can, Cognar has a record of the stories and of the differences in stories.

And maybe there's really kind of two subsets in this group or three or four subsets in this group.

And, and each subset views the situation in a very different way.

Well, at the very least Cognar is going to help the group to understand that there's, you know, different ways to look at this problem and different proposed solutions.

And then it's going to record the, the, the evolving stories that are related to the, to this, you know, like how they, how a particular story started and what it, what it grew to become over the course of cognition.

And so, you know, maybe that, maybe the group will actually come to a, a well, you know, widely accepted decision on some topic, or maybe it won't, but at least it's going to have a functional process to, you know, to, to go through, to, to understand itself.

Right.

So that's the, that's the, that's the, the, hopefully yes.

And, but you know, that what happens, the dynamics depends on what the group is and what the topic is.

Thank you.

All right.

Next question from Abbas Tolgate, who wrote two questions.

One, how does Cognar maintain consistency in story graph updates when passages are the single source of record?

Second question, how are changes propagated across the story graph and associated models without conflicts?

Can you, can you flip back in the slides to the, to the process, process of creating the story?

It's kind of far back.

That one more.

Yeah, there we go.

Oh, this one.

Yeah.

There you go.

Thank you.

Okay.

So, so, so one could, you know, we're, we're, we're thinking about how should Cognar be designed, you know, how should, how should it work?

You know, what should it do?

And, you know, one could imagine a user could input text, type in something, and then that text becomes a story graph fragment here.

And then the story graph fragment becomes a story graph.

And then the re and then maybe the user realizes, oh, I didn't, I didn't mean that, or I didn't make that clear.

I want to add something.

So how should the user edit this thing?

Right?

You could edit the story graph or you could edit the graph fragment, or you could edit the original text and then rebuild the story from there.

Right?

So how should it happen?

Well, the simplest approach and the approach I suggest in the, in the paper is that all the editing really happened on the text passages, because if, if you allow edits on the story graph, then they have to propagate in this diagram, have to propagate upward to, so, you know, to change the original text passages so that everything is in agreement, all the different sources of information are in agreement.

So it would be easier and simplest if there's only one source of information, and that the entire graph can be, you know, regenerated from that source as the, as the information changes.

What was the second question?

I think maybe I answered.

How, how is the internal consistency and, and the statistical consistency, and then also the, the changes of through actual propagation, how does conflict, how are changes propagated across the story graph and

associated models without conflicts?

Yeah.

Yeah.

That's, that's, that's, that's, I think I answered the question, or at least I tried to.

Um, so there's only one source of record and that's the text passages itself.

But, uh, suppose now that the, that the system, uh, suppose the user created a story and the system, uh, uh, responds and says, Hey, I don't, I don't understand is, is, you know, is Bob, Bill's brother or his boss?

I don't understand, you know, what this relationship is or who, when you say he, which he are referring to Bob or Bill, or, you know, who, who, who is it that this, this passage refers to?

So now the, now you have a situation where, um, you know, maybe, maybe the question is, is, uh, are they brothers or are they coworkers?

Maybe that's the question that this imposes.

And maybe the user responds, they're brothers.

And you know, as a response, well, how does that, how does that response, you know, get into the story graph, right?

Because you don't want the system to edit the story graph because then the system would have to change, you know, propagate that changed information back up into the text passages.

And that might not work so well.

You know, what if the system doesn't propagate it correctly?

Uh, the user could simply change their original text passages and, and make clear that what the relationship is, you know, whatever the question is.

So, you know, that's a clear way.

The user just changes the text passage itself.

But another way is there's some, there's metadata associated with a text passage and that metadata might actually be, you know, include the kind of the conversation between the user and the, and the, and the system.

The system says, the system asks, you know, are they brothers or coworkers?

And the user's responses, their coworkers.

Okay.

Well that that's meta information that is now, you know, saved as meta information and, uh, can be, become part of the graph, you know, downstream.

What about the human element of conflict?

Like using code as law, legal code, plain text repositories, versioning and forking and instancing.

How does the social reality of the conflict not go into the fractal Linux kernel separation and what would actually, you said one version of record and whether or not that resolves logical ambiguity, that's an internal question.

What about the actual existence and production and power and influence over that one version of record if there really is one?

Um, are you asking, uh, how reliable is a versioning system?

Uh, like if there's some kind of versioning system where the user, you know, changes information, is there a reliable versioning system that can keep track of all these changes?

Is that?

Yeah.

Given a reliable, complete, perfect, get type versioning of the text.

How, how does that not, or does it, or et cetera, have conflict outside of the text?

Well, I, I, I'm not exactly sure what you're asking, but let me pose this situation to you.

Suppose that, um, I don't know, uh, um, suppose that the, that the system is, um, what the story is saying and ask the user, you know, are, uh, is bills, uh, these two people related or they coworkers.

And the user responds, the user makes a response.

So therefore changes the record changes the metadata and it says something really vague.

Like, uh, they're, they, they, they, they know each other.

That's the response.

That didn't really address the question.

Like it didn't resolve the issue that was raised.

Right.

But then, so what that means is that the story is simply vague.

Right.

Right.

So the story is not a, maybe not a great story.

It's just has inconsistencies within it.

It's, it's, uh, not logical maybe, or it's just too vague to understand what, what is being said.

So that's not the, that's not the problem with, it's not a problem of Cognar itself.

That's a problem with the telling of the story and its completeness and its logical consistency and things like that.

And just like any, just like, just if you and I were in a room together and we're hashing out what we think about some situation, I may, I might make some vague statement to you about, you know, people involved and you might think in your head, what, what is he saying?

And you know, like, that's, that's the way it is, right?

That's the way it goes.

And you might ask me, you know, can you clarify that, but maybe I give you some unclear answer and then you still don't know what I'm saying.

And you walk away thinking, well, John doesn't really know what he's saying, but that's the, that's also can happen here.

So if I'm a user and there's some, a group that I'm participating in, and there's some event that I'm offering a

story to.

And I, and I submit some, a story that is, especially if I have a history of submitting stories that are unclear, logically inconsistent, incomplete.

You know, my reputation in that group is probably not going to rise very high.

You know, people will eventually figure out that he, he doesn't, he's not clear on what he's saying, you know, or else he doesn't know what he's saying.

Yeah.

So, you know, those are, that's, that's part of the dynamics, interpersonal dynamics and, and the, you know, assessing, assessing who in the group can be relied on to give a thoughtful, you know, reasonably useful contribution.

And who should we pay attention to? Like the group has an attention mechanism, right?

Naturally, and maybe even facilitated by Cognar to some degree.

And that attention mechanism would sort of naturally focuses on contributions from people who are known to provide clear, useful information in the past, or maybe people who have good predictive capacity.

Like maybe I get a record, a record and a reputation because I've participated in this group many times or on many different events.

And I, a lot of what I said was going to happen in the world actually happened.

You know, I'm good at making forecasts of what's in, maybe, maybe I'm good at making forecasts.

So maybe the group is going to weigh my contribution a little bit more than some other person's contribution that was unclear or made poor forecasts.

Maybe I, maybe I kind of didn't answer your question, Daniel, but.

Yeah.

I think it, it touches on, and you cite and discuss some related projects related to like meta governance.

Like you talk about modular policy and about attention economics.

And then also the trust networks.

It's a lot of interesting work, like John Ash's work in cognicism and just thinking about how the attention distribution, which is a lot of interesting work.

And then the attention distribution, which is changing moment to moment based upon what kinds of message passing and, and quantum reference frames the interpretant deploys in that moment in the information environment setting.

And then how that relates to like formal and informal and all of these heterogeneous systems, which I think will bring us through the series towards active inference.

And what does active inference do beyond you saying there's positive alternative narratives?

However, such is some of the most common statements that people implicitly or explicitly say.

Yeah.

Yeah.

Well, you have, you raised many good points.

I'd like to talk to, to speak to a couple of them.

One is that, you know, by, by, by thinking of, you know, group decision-making as a cognitive process and thinking of a cognitive process as being a purposeful process.

And, and, and, and using, using active inference and other frameworks to potentially, you know, evaluate how, maybe there's ways to evaluate how functional a given process was.

Maybe in retrospect, a group can look back at its history and assess how functional is our cognition.

How, how, how are we doing?

How, how is information flowing?

How, what are the many, you know, from, from a scientific standpoint, what are the many ways that we could evaluate the process of cognition for its, you know, capacity or functionality or other things?

So, you know, those are some of the nitty gritty scientific questions that, that this project is sort of offering in a way.

Another thing that you mentioned that I'm really glad you mentioned was meta governance.

And it's an interesting group and I, I, you know, cite them in the first paper at the end.

But I did, I did, it reminded me that I wanted to say that, that, you know, Cognar itself is sort of a group agnostic.

Like different kinds of groups could use Cognar, Cognar app, and that, that, the different groups might have different, you know, levels of formality or different ways they want to make decisions.

Maybe if it's a decision-making, you know, process they're interested in, maybe, maybe they want to use majority vote.

Maybe they want to use rank choice voting.

Maybe they have some other, you know, unusual system they want to use.

Maybe they want, maybe the group is organized in a particular way.

Maybe it's an autocratic group or, you know, or a very democratic group or something, right?

So Cognar could be used by any group.

The group is going to define what it is and how it works and what its rules are and, you know, how things go.

And then Cognar seeks to be the structure on which then, you know, cognitive event, cognitive group events can occur.

Right. And that is really good because it doesn't, Cognar is not responsible for, you know, forcing all users to, to use some particular method or approach or rule set or whatever.

You know, there's room for innovation within the Cognar system.

There's room for experimental experiment, experiment, experiment within the Cognar system.

There's a history of, of events that can be assessed in hindsight for how well they worked.

How did they, how well did they guide a group towards shared, you know, widely, widely shared and valued solutions to problems and things like that.

So, yeah.

Awesome. Okay.

Another question from live chat and anyone else can continue to write questions and we can continue to explore a few things.

Synthetic data generator wrote, can it be used as a protocol for group mental model harmonization?

Well, that's what it is. You know, if I understand the question correctly, that's sort of what it is.

If you and I are together in a room and we're talking about some situation in the world, I have a model of, in

my mind, about how the world works and what will happen if we do something or not do something.

And what it means and what it means and what it's, what the future might be.

And you have the same, you have a, your own model in your head and we're, and we're communicating aspects of that.

But what, but what Cognar does is it kind of makes that process available to the large group setting where there's potentially millions of people who are participating in this.

And what it, what it, what it does is it makes the model explicit, right?

It, it puts it down in an explicit way so that it can be assessed and analyzed and considered as whether it's different, how different it is from other ways of approaching the situation or, you know, what its components are.

So, you know, I, I, I, that is what Cognar does is it, it, it represents a user's mental model of, of a situation.

Yeah.

Yeah.

And I think an important aspect of the harmonization is whether or not any given sense of harmonization is achieved or whether that's like a positive attribute used like coherence informally by taking like a dynamical systems, parametric modeling, Bayesian probabilistic.

There actually are ways like using fMRI analysis on the brain that rhythms of different timescales, different mutual information, different KL divergences.

Like there's a whole toolkit of harmonics and analysis.

So this is one mapping strategy and representation.

However, it doesn't need to be the only mapping or representation, one that has some great causal properties though, and helps be like kind of a repository format for the perception, cognition, and action that make things about a given system specific rather than like the generic active inference formalism.

Yeah.

There's a, recently there's been several papers published on how groups harmonize their belief systems as they come together to, you know, understand the world or make decisions.

So there's a, already prior work on what kind of, what harmonization in a group setting might look like and how that works and how it might be even mathematically approached, you know, for modeling purposes.

Yeah.

Yeah.

And another, another thing that you just raised there a moment ago, Daniel, is that, is the idea of a repository.

So suppose that Cognar existed and there were, you know, say millions of stories in Cognar and, you know, hundreds of thousands of group events that could be analyzed in retrospect.

Think of the value that would be to science for understanding cognition and understanding, you know, how information is conveyed and used and flows and, you know,

what kind of designs for rule systems or inputs or whatever facilitate cognition in the best way.

I mean, just the, just the existence of a repository of externalized belief models of, you know, complex externalized belief models of a variety of individuals on a variety of topics itself would be extremely valuable for cognitive science and computer science and various other, you know, psychology, various other aspects of science.

Yeah.

Yeah.

This relates to a question that synthetic data generator followed up on.

It's an interesting question.

Larkin and Simon were the seminal researchers in this diagrammatic reasoning.

Is the explicit model visual or linguistic?

Is the, if I can interpret that, the question, is the story graph visual or linguistic?

Is that about it?

So it's, it's a, the story graph, one of the purposes of the story graph, the meaning representation is to allow computation.

So, um, in that sense, I guess you would say it's linguistic.

It's, it's, it's trying to, um, record the, trying to capture the meaning of what, you know, a phrase is what the user is trying to say.

Right.

And how that phrase relates to other phrases of what, what the user is trying to say.

But it's also visual in the sense that suppose, and I'm glad you brought the slide up, but suppose I'm, I, I, I, I'm creating a story and now I want to check whether the system is understanding that story correctly.

Right.

So what I might want to do is I might want to view a portion of the story graph under the hood and see if the system put those things together correctly.

So that aspect could be sort of visual as a user.

I could look at something, you know, like this, like what you have on the screen and go like, okay, I see that, uh, I see that these two concepts were properly connected with each other.

It actually, you know, Sue actually is related to Robert and, uh, you know, this particular way, or it was, it was Sue that dropped the gun, not Bob or, you know, whatever the story is.

So, uh, I use, there might be visual tools that the user would have available in order for them to, you know, query the, query the underlying story graphs, or even query someone else's story graph.

You know, there might be visual tools for that and, and, and a result from a query might be a visual result so that, uh, you know, it sort of makes sense to how different elements are connected with each other.

But the, but the, but the, you know, kind of the core thing happening under the hood is computation, uh, based on the linguistic meaning of, uh, phrases and, and the intent of the user.

Yeah.

Yeah.

A lot there, especially programs considered and graphs considered as diagrams, like from an applied category theoretic perspective, the data structures themselves already being diagrammatic, symbolic.

Now they, they might not be the right semantic representations for given system with this available, observable, with this unobservable model for this person with this accessibility, that language, like, et cetera.

But that is a cognitive situation with the human computer interface.

So whether it's a phone and uses this interface or uses another interface, part of the interesting question with a generative model interoperability is can those different blankets, different screens be also connected to a

system that has an underlying representation that those perspectives come filtered through.

Right, right, right, right, right, right.

You, you, you raised, uh, um, the field of, uh, applied category theory there a minute ago, which is really interesting.

Lots of applications of applied category to this sort of setting, because maybe you want to translate this graph into some other representation.

And you want to make, and you want to, uh, you want that, uh, that, um, translation to preserve meaning.

Well, that, you know, that, that's a functor in, in, um, in applied category theory.

So there might be ways to use applied category theory and, in, in even designing the structure of this, you know, the under the kind of like what structurally, how does a story graph work out?

What is it, how is it stored?

How is it retrieved from a database and, you know, how is it manipulated and how is it, you know, what kind of functors translate a particular story into a different version or visualization or, you know, what kind of functors translate a particular story into a different version or visualization or.

or, you know, uh, uh, uh, uh, but really just a different, you, uh, different version of that story or, uh, sub, sub story or translated from a story graph into English or a story graph into animation or, you know, like whatever the case, whatever the case might be.

Awesome.

Well, this was a great first go for dot one.

Um, maybe just restate what are we going to head towards and dot to what would be fun and what, what, for those who are watching now could do between now and then.

Yeah.

Yeah.

Well, um, uh, so we've, today we've kind of talked about the kind of the general high level.

What is Cognar and what is it supposed to do?

Uh, next time when we get together, we'll talk more about meaning representations and, you know, what, what kind of, you know, what should the Cognar meeting representation look like?

What should it do?

What, what characteristics should it have?

How was it put together?

How does, maybe we'll have a chance to talk about how does applied category fit into all of this?

And, um, um, in the meantime, uh, you know, I, I encourage, uh, watchers to take a look at the papers and, uh, and also if there's questions, um, you know, feel free to send, send questions.

I think I can be reached through the Institute somehow perhaps, and, um, or post questions and maybe we can try to pick up any of those questions or comments or suggestions, uh, you know, uh, that are offered in the intermediate, uh, intermediate part.

Pick it up next time.

Okay.

Awesome.

Thank you again, John and Lee too.

So see you next time.

Sounds good.

Bye.

Bye.

Bye.

Bye.